

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (RI) (Credit System) Degree Examinations**

April 2023

21RI304 - ALGORITHM AND PROGRAMMING CONCEPTS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain looping constructs with flowchart.	5	L*2	1	1
	b) Explain standard data types used in python with example.	5	L2	1	1
	c) Explain type conversions types with an example.	5	L2	2	2
	d) Define tuple and dictionary, illustrate with a program.	5	L2	2	2
2.	a) List out factors, reason why python is considered preferred programming language and its applications. Explain.	10	L1	1	1
	b) Explain control flow types in detail with flowchart and example.	10	L2	2	2
3.	a) How could python interpreter be used?	5	L1	1	1
	b) List out applications and installation steps of python.	5	L1	1	1
	c) Define functions. Explain its types, flow of execution with an example in detail.	10	L2	2	2
Unit – II					
4.	a) Define OOPS. Explain principles of object oriented programming.	5	L1	3	1
	b) Explain inheritance with example.	5	L2	3	2
	c) Explain event driven programming with flowchart.	5	L2	4	2
	d) Explain future classes in asyncio module.	5	L2	4	2
5.	a) What are virtual functions? Explain in detail.	5	L1	3	1
	b) Define package. Explain with a program.	5	L2	3	1
	c) Explain concept of coroutines in asyncio module.	10	L2	4	2
6.	a) Explain encapsulation and polymorphism with example.	10	L2	3	2
	b) Explain event loops.	5	L2	4	1
	c) Explain transports classes provided in the asyncio module.	5	L2	4	2
Unit – III					
7.	a) Define data structures. Explain classification of data structures.	5	L2	5	2
	b) What is algorithm? Explain notion along with characteristics of an algorithm.	5	L2	5	2
	c) Explain asymptotic notations with its properties.	5	L2	5	2
	d) Illustrate quick sort with algorithm for this example. 17,9,22,31,7,12,10,21,13,29,18,20,11	5	L3	5	2
8.	a) Explain fundamentals of problem solving and analysis of efficiency of algorithm.	10	L2	5	2
	b) Explain binary search with example in detail	10	L3	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
